

Week1 Homework

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Prob 1

a. Self-supervised learning is for data that has no label, we find a way to generate the label for all data, and let model to learn. E.g. For LLM, we can delete the last word of a text, and let model to predict the last word.

Different to unsupervised learning, self-supervised learning **do** has label for each data.

b. The research group use unsupervised learning to solve predicting image rotations, and gets a best result in all unsupervised learning, with similar accuracy to supervised learning. The group use this method in other problems, getting a result as significant as the current problem.

c. Nowadays embodied AI can't feel tactile sensing very accurately and comprehensively, so they can't react the physical world in time.

To solve this problem, we may make the hand of robot to be soft and be filled by liquid, thus we can measure the pressure to deduce the force that every point is currently act.

Prob 2

We have $n = 15$, $\mu = 6$, so $\Pr(X \geq (1 + \frac{2}{3})\mu = 10) \leq e^{-\frac{8}{9}} \approx 0.41$.

Prob 3

a. i. to ii.: We have

$$\begin{aligned} f(w') - f(w) - \langle \nabla f(w), w' - w \rangle &= \int_0^1 \langle \nabla f(w + u(w' - w)) - \nabla f(w), w' - w \rangle du \\ &\leq \int_0^1 Lu \|w' - w\| \|w' - w\| du \\ &= \frac{L}{2} \|w' - w\|^2. \end{aligned}$$

ii. to i. (helped by chatgpt): We have

$$\max(f(w') - f(w) + \langle \nabla f(w), w - w' \rangle, f(w) - f(w') + \langle \nabla f(w), w' - w \rangle) \leq \frac{L}{2} \|w - w'\|^2,$$

likely,

$$\max(f(w) - f(w') + \langle \nabla f(w'), w' - w \rangle, f(w') - f(w) + \langle \nabla f(w'), w - w' \rangle) \leq \frac{L}{2} \|w' - w\|^2,$$

so

$$|\langle \nabla f(w), w - w' \rangle + \langle \nabla f(w'), w' - w \rangle| \leq L \|w - w'\|^2,$$

that is

$$|\langle \nabla f(w) - \nabla f(w'), w - w' \rangle| \leq L \|w - w'\|^2.$$

Let $w' = w + th$, we have

$$|\langle \frac{\nabla f(w + th) - \nabla f(w)}{t}, h \rangle| \leq L \|h\|^2.$$

Let $t \rightarrow 0$, we have

$$|\langle H(w)h, h \rangle| \leq L \|h\|^2,$$

that is

$$|h^t H(w)h| \leq L \|h\|^2,$$

so

$$\|H(w)\| \leq L.$$

Then we have

$$\|\nabla f(w + h) - \nabla f(w)\| \sim \|H(w)h\| \leq L \|h\|,$$

integral, we have

$$\|\nabla f(w) - \nabla f(w')\| \leq L \|w - w'\|.$$

b. i. $\rightarrow$ ii.: Let $z = ax + (1 - a)y$, we have

$$\begin{aligned} a(f(z) - f(x)) &\leq (1 - a)(f(y) - f(z)) - a(1 - a)\frac{\mu}{2}\|x - y\|^2, \\ \frac{f(z) - f(x)}{(1 - a)\|x - y\|} &\leq \frac{f(y) - f(z)}{a\|x - y\|} - \frac{\mu}{2}\|x - y\|, \\ \frac{f(z) - f(x)}{\|z - x\|} &\leq \frac{f(y) - f(z)}{\|y - z\|} - \frac{\mu}{2}\|y - x\|. \end{aligned}$$

Let $a \rightarrow 1$, that is $z \rightarrow x$, we have

$$\begin{aligned}\langle \nabla f(x), \frac{y-x}{\|y-x\|} \rangle &\leq \frac{f(y)-f(x)}{\|y-x\|} - \frac{\mu}{2}\|y-x\|, \\ f(y) &\geq f(x) + \langle \nabla f(x), y-x \rangle + \frac{\mu}{2}\|x-y\|^2.\end{aligned}$$

ii. $\rightarrow$ iii.: We have

$$f(y) - f(x) \geq \langle \nabla f(x), y-x \rangle + \frac{\mu}{2}\|x-y\|^2$$

and

$$f(x) - f(y) \geq \langle \nabla f(y), x-y \rangle + \frac{\mu}{2}\|y-x\|^2,$$

so we have

$$0 \geq \langle \nabla f(x) - \nabla f(y), y-x \rangle + \mu\|x-y\|^2,$$

that is

$$\langle \nabla f(x) - \nabla f(y), x-y \rangle \geq \mu\|x-y\|^2.$$

iii. $\rightarrow$ iv.: Let $y = x + th$, we have

$$\left\langle \frac{\nabla f(x+th) - \nabla f(x)}{t}, h \right\rangle \geq \mu\|h\|^2.$$

Let $t \rightarrow 0$, we have

$$\begin{aligned}\langle H(x)h, h \rangle &\geq \mu\|h\|^2 \\ h^t H(x)h &\geq \mu\|h\|^2 \\ \lambda_{\min} H(x) &\geq \mu.\end{aligned}$$

iv. $\rightarrow$ ii. (helped by chatgpt): Let $g(a) = f(ax + (1-a)y)$.

We have $g'(a) = \langle \nabla f(ax + (1-a)y), x-y \rangle$, $g''(a) = (x-y)^t H(ax + (1-a)y)(x-y) \geq \mu\|x-y\|^2$.

So $g'(a) \leq g'(1) - (1-a)\mu\|x-y\|^2$, $g(1) - g(0) \leq g'(1) - \frac{\mu}{2}\|x-y\|^2$.

That is $f(y) \geq f(x) + \langle \nabla f(x), y-x \rangle + \frac{\mu}{2}\|x-y\|^2$.

ii. $\rightarrow$ i. Let $z = ax + (1-a)y$, we have

$$\begin{aligned}
\frac{f(x) - f(z)}{(1-a)\|x-y\|} + \frac{f(y) - f(z)}{a\|x-y\|} &= \frac{f(x) - f(z)}{\|x-z\|} + \frac{f(y) - f(z)}{\|y-z\|} \\
&\geq \frac{\langle \nabla f(z), x-z \rangle + \frac{\mu}{2}\|x-z\|^2}{\|x-z\|} + \frac{\langle \nabla f(z), y-z \rangle + \frac{\mu}{2}\|y-z\|^2}{\|y-z\|} \\
&= \langle \nabla f(z), \frac{x-z}{\|x-z\|} \rangle + \frac{\mu}{2}\|x-z\| + \langle \nabla f(z), \frac{y-z}{\|y-z\|} \rangle + \frac{\mu}{2}\|y-z\| \\
&= \frac{\mu}{2}\|x-y\|.
\end{aligned}$$

That is

$$a(f(x) - f(z)) + (1-a)(f(y) - f(z)) \geq a(1-a)\frac{\mu}{2}\|x-y\|^2,$$

so

$$f(ax + (1-a)y) \leq af(x) + (1-a)f(y) - a(1-a)\frac{\mu}{2}\|x-y\|^2.$$

Prob 4 (use chatgpt to understand the goal of problem)

a. Let $x_{n+1} = x_n - \eta \nabla f(x_n)$ where $\eta \leq \frac{1}{L}$. Let $f^* = \inf f$. We have

$$\begin{aligned}
f(x_{n+1}) &\leq f(x_n) + \langle \nabla f(x_n), -\eta \nabla f(x_n) \rangle + \frac{L}{2}\|\eta \nabla f(x_n)\|^2 \\
&= f(x_n) - (\eta - \frac{L\eta^2}{2})\|\nabla f(x_n)\|^2 \\
&\leq f(x_n) - \frac{\eta}{2}\|\nabla f(x_n)\|^2.
\end{aligned}$$

So

$$\begin{aligned}
f(x_{n+1}) &\leq f(x_n) - \frac{\eta}{2}\|\nabla f(x_n)\|^2 \\
&\leq f(x_{n-1}) - \frac{\eta}{2}(\|\nabla f(x_n)\|^2 + \|\nabla f(x_{n-1})\|^2) \\
&\dots \\
&\leq f(x_0) - \frac{\eta}{2} \sum_{i=0}^n \|\nabla f(x_i)\|^2.
\end{aligned}$$

Thus

$$\sum_{i=0}^n \|\nabla f(x_i)\|^2 \leq \frac{2}{\eta}(f(x_0) - f(x_{n+1})) \leq \frac{2}{\eta}(f(x_0) - f^*).$$

$$\min_{i \in [0, n]} \|\nabla f(x_i)\|^2 \leq \frac{2(f(x_0) - f^*)}{\eta(n+1)}.$$

So $\|\nabla f\|^2$ converges to 0 at speed of $O(\frac{1}{T})$.

b. Let $\eta \leq \min(\frac{1}{L}, \frac{1}{2\mu})$, we have

$$\begin{aligned}
f(x_{n+1}) &\leq f(x_n) - \frac{\eta}{2} \|\nabla f(x_n)\|^2 \\
&\leq f(x^*) + \langle \nabla f(x_n), x_n - x^* \rangle - \frac{\mu}{2} \|x_n - x^*\|^2 - \frac{\eta}{2} \|\nabla f(x_n)\|^2 \\
&= f(x^*) - \frac{1}{\eta} \langle x_{n+1} - x_n, x_n - x^* \rangle - \frac{1}{2\eta} \|x_{n+1} - x_n\|^2 - \frac{\mu}{2} \|x_n - x^*\|^2 \\
&= f(x^*) - \frac{1}{2\eta} \|x_{n+1} - x^*\|^2 + \frac{1}{2\eta} \|x_n - x^*\|^2 - \frac{\mu}{2} \|x_n - x^*\|^2 \\
&= f(x^*) + \frac{1 - \mu\eta}{2\eta} \|x_n - x^*\|^2 - \frac{1}{2\eta} \|x_{n+1} - x^*\|^2
\end{aligned}$$

That is

$$\begin{aligned}
f(x_{n+1}) - f(x^*) &\leq \frac{1 - \mu\eta}{2\eta} \|x_n - x^*\|^2 - \frac{1}{2\eta} \|x_{n+1} - x^*\|^2, \\
\frac{f(x_{n+1}) - f(x^*)}{(1 - \mu\eta)^n} &\leq \frac{1}{2\eta(1 - \mu\eta)^{n-1}} \|x_n - x^*\|^2 - \frac{1}{2\eta(1 - \mu\eta)^n} \|x_{n+1} - x^*\|^2, \\
\sum_{i=0}^{n-1} \frac{f(x_{i+1}) - f(x^*)}{(1 - \mu\eta)^i} &\leq \frac{1 - \mu\eta}{2\eta} \|x_0 - x^*\|^2 - \frac{1}{2\eta(1 - \mu\eta)^{n-1}} \|x_n - x^*\|^2, \\
f(x_n) - f(x^*) &\leq \frac{(1 - \mu\eta)^n}{2\eta} \|x_0 - x^*\|^2.
\end{aligned}$$

We have $1 - \mu\eta \geq 1 - \mu\frac{1}{2\mu} = \frac{1}{2}$, so $\frac{1}{2} \leq 1 - \mu\eta < 1$. Thus $f(x_n) \sim O((1 - \epsilon)^n)$.